

Supplementary Material for “Auditing Cross-Dataset Claims in Wearable EMG Representation Learning: A Public-Data Case Study”

Anonymous Authors

I. SUPPLEMENTARY SCOPE

This supplement provides additional details that support the main manuscript without changing its claim boundary. The central claim remains audit-gated: representation-level signals are reported only when their semantic, diagnostic, downstream, and artifact gates are satisfied. The supplementary tables expand the related-work positioning, source-data ledger, domain-identifiability diagnostics, ontology gate, downstream baseline checks, and descriptive proxy analyses.

II. RELATED-WORK CLAIM MAP

III. REPRODUCIBILITY LEDGER

IV. ALIGNMENT SENSITIVITY CHECKS

The released source tables retain direction-labeled source-target rows because the original sampling and reference-cluster bookkeeping were directional. Linear CKA itself is symmetric for a fixed pair of embedding matrices, so the claim-bearing summary uses reciprocal-averaged unordered pairs and leave-one-domain checks.

These sensitivity tables support only a small unwhitened geometry shift. They do not establish whitening-invariant alignment, calibrated domain invariance, supervised semantic transfer, or downstream control benefit.

V. SIX-DOMAIN FIXED-PROBE DIAGNOSTICS

These are independent fixed-probe diagnostics; they do not reconstruct the original adversarial-head confusion matrix, true or predicted marginals, fixed-set entropy, balanced accuracy, mutual information, optimal-relabel accuracy, or readout-training trace.

For HARMONIC, the rebuilt six-packaged-domain probe used a reconstructed public text/probability stream rather than raw HARMONIC EMG. Perfect recall therefore shows package-stream recoverability, not raw-EMG physiological separability.

VI. ONTOLOGY GATE

VII. DOWNSTREAM BASELINE SENSITIVITY

VIII. DESCRIPTIVE PROXY ANALYSES

IX. ADDITIONAL PROVENANCE AND DIAGNOSTIC MATERIAL MOVED FROM MAIN TEXT

The following ledgers and diagnostic figures were moved from the main manuscript to reduce main-text density while preserving reviewer access to the full audit trail.

X. ARCHIVE BOUNDARY

The review source package includes manuscript-ready CSV/JSON summaries, figures, checksums, and this supplement. It intentionally excludes raw third-party EMG data, model checkpoints, embedding caches, raw-window feature parquet files, trial-level prediction caches, local launcher logs, machine-local paths, and public repository or archive identifiers. The package supplied through the review system is self-contained and includes regenerated PDFs, source data, figures, manifest, and checksum ledger.

TABLE I
POSITIONING RELATIVE TO COMMON CROSS-DOMAIN EMG STUDY FAMILIES

Study family	Typical claim	What the claim does not guarantee	Role of this paper
Self-supervised EMG pretraining	Reusable features or improved sample efficiency	Task-valid cross-dataset supervised transfer or superiority over strong raw-feature baselines	Tests downstream claims as explicit gates rather than assuming transfer from representation learning alone
Domain-adversarial EMG	Lower dataset or subject identifiability	Calibrated invariance, optimal-relabel non-recoverability, or preservation of task information	Treats below-reference dataset-ID accuracy as an audit failure unless supported by balanced, relabeled, and probe diagnostics
OOD benchmark studies	Reported cross-subject or cross-session performance	Ontology validity across arbitrary corpora with different task definitions	Separates four-domain geometry, six-packaged-label domain-ID, and task-valid supervised endpoints
Large-corpus foundation-model EMG or	Aspirational reuse across devices and tasks	Clinically meaningful control transfer, fairness, or reduced recalibration burden	Requires downstream and ontology gates before deployment-facing language is allowed
Sensor/statistical alignment methods	Reduced feature-distribution mismatch	Shared semantic control intent or user-facing performance benefit	Treats geometry and ontology as separate prerequisites for transfer claims

TABLE II
SOURCE-DATA LEDGER FOR MAIN QUANTITATIVE CLAIMS

Claim or endpoint	Source-data folder	Unit or split	Decision boundary
Training mixture counts	Canonical release tables	Packaged records	Counts are package records, not full raw-dataset sizes
Label-agnostic CKA shift	Canonical release tables; alignment sensitivity	Six unordered four-domain pair means plus direction-labeled source rows	Bounded pass only for small unwhitened geometry shift
Whitened CKA result	Canonical release tables; alignment sensitivity	Six unordered four-domain pair means plus direction-labeled source rows	Fail for whitening-invariant alignment
Canonical adversarial dataset-ID endpoint	Canonical release tables	Logged mini-batch readout	Failed calibration diagnostic, not invariance evidence
Six-packaged-domain fixed probe	Six-packaged-domain fixed-probe folder	5 encoders $\times$ 10 probe seeds	Reject calibrated invariance for probed embeddings
Ontology gate	Ontology controls	Predeclared control-intent mapping	Hyser-GRABMyo transfer not evaluable
Downstream baseline gates	Canonical baselines; post-freeze confirmation audit	Same-subject and held-out-subject units	Classical signals can help, but pretrained feature superiority fails
Audit integrity	Audit checks	Frozen package checks	Supports reproducibility, not deployment readiness

TABLE III
UNORDERED FOUR-DOMAIN CKA SENSITIVITY

Unordered pair	Mean delta	Minimum direction	Maximum direction	Interpretation
HARMONIC-NinaPro/MeganePro	+0.00416	+0.00237	+0.00595	Positive in both direction-labeled source rows
HARMONIC-GRABMyo	+0.00491	-0.00186	+0.01168	Positive pair mean, but one direction-labeled row was negative
HARMONIC-Hyser	+0.00244	+0.00184	+0.00303	Positive in both direction-labeled source rows
NinaPro/MeganePro-GRABMyo	+0.00663	+0.00231	+0.01095	Positive in both direction-labeled source rows
NinaPro/MeganePro-Hyser	+0.00220	+0.00200	+0.00241	Positive in both direction-labeled source rows
GRABMyo-Hyser	+0.00305	+0.00271	+0.00339	Positive in both direction-labeled source rows

TABLE IV
LEAVE-ONE-DOMAIN CKA SENSITIVITY

Left-out domain	Mean remaining delta	Minimum direction	Maximum direction	Direction-row sign count
HARMONIC	+0.00396	+0.00200	+0.01095	6 positive, 0 negative
NinaPro/MeganePro	+0.00347	-0.00186	+0.01168	5 positive, 1 negative
GRABMyo	+0.00293	+0.00184	+0.00595	6 positive, 0 negative
Hyser	+0.00523	-0.00186	+0.01168	5 positive, 1 negative

TABLE V
SIX-PACKAGED-DOMAIN FIXED-EMBEDDING PROBE SUMMARY

Condition	Encoders	Probe seeds/encoder	Classes	Accuracy	Balanced accuracy	Macro-F1	Interpretation
True package labels	5	10	6	0.92748	0.92262	0.92086	Package/source identity remained highly recoverable from frozen embeddings
Shuffled training-label control	5	10	6	0.18944	0.18829	0.18446	Control remained near the six-class reference and majority baseline

TABLE VI
PER-DOMAIN RECALL IN THE SIX-PACKAGED-DOMAIN FIXED PROBE

Domain	True support mean	Predicted support mean	Recall mean	Recall range	Interpretation
HARMONIC	1291.6	1293.22	1.00000	1.00000–1.00000	Package stream was perfectly recoverable; see boundary below
NinaPro DB10/MeganePro	1249.9	1249.98	0.99962	0.99759–1.00000	Primary feature-bank domain was nearly perfectly recoverable
PhysioNet GRABMyo	1270.9	1271.40	0.99972	0.99841–1.00000	Primary feature-bank domain was nearly perfectly recoverable
PhysioNet Hyser	1247.3	1257.76	0.99992	0.99920–1.00000	Primary feature-bank domain was nearly perfectly recoverable
emg2pose	1333.3	1267.70	0.77639	0.69685–0.86464	Limited indexed subset; lower recoverability than primary domains
emg2qwerty	966.8	1019.74	0.76007	0.51157–0.88505	Limited indexed subset; lower recoverability than primary domains

TABLE VII
ONTOLOGY-GATE DECISIONS

Endpoint	Source	Target	Accepted classes	Allowed?	Permitted wording
GRABMyo same-dataset session shift	GRABMyo	GRABMyo	17	Yes	Same-dataset task-valid screen
Hyser same-dataset session shift	Hyser	Hyser	34	Yes	Same-dataset task-valid screen
Hyser to GRABMyo supervised transfer	Hyser	GRABMyo	0	No	Not evaluable; no documented one-to-one control-intent mapping exposed by frozen package
GRABMyo to Hyser supervised transfer	GRABMyo	Hyser	0	No	Not evaluable; no documented one-to-one control-intent mapping exposed by frozen package

TABLE VIII
DOWNSTREAM BASELINE CHECKS USED AS CLAIM-INCLUSION GATES

Screen	Comparison	Effect	Gate interpretation
Same-subject frequency comparator	Raw + pretrained vs. raw + frequency	−0.03516 normalized log-AULC	Conventional frequency comparator did not rescue a pretrained-feature advantage
Same-subject frequency comparator	Frequency-only vs. raw statistics	−0.00425 normalized log-AULC	Frequency features were close to raw statistics but did not beat them overall
Held-out-subject classical signal comparator	Classical + pretrained vs. classical	+0.00605 normalized log-AULC; sign-flip $p = 0.16242$	Below effect threshold, dataset-heterogeneous, and failed positive gate
Held-out-subject classical signal positive control	Raw + classical signal features vs. raw statistics	+0.0837 normalized log-AULC in main manuscript	Offline GRABMyo/Hyser ridge-readout screen detected useful classical signal-feature information under this constrained benchmark; not an online-control, EMGBench, or state-of-the-art baseline-suite result

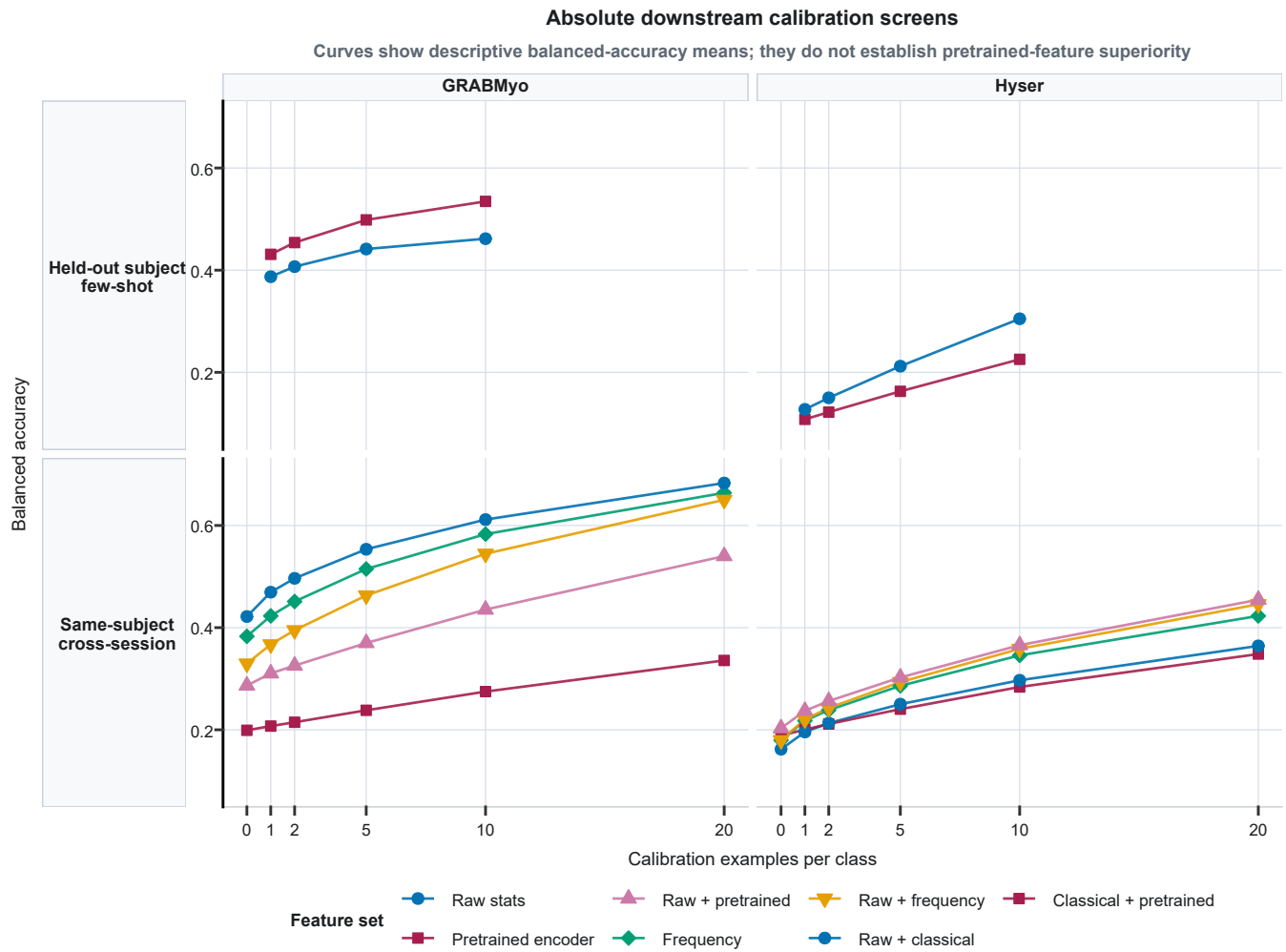


Fig. S1. Absolute downstream balanced-accuracy curves for the claim-inclusion screens. Panels show absolute accuracy across calibration counts rather than only normalized log-AULC deltas. The curves confirm that pretrained features did not establish superiority over raw statistics or classical signal features under these offline ridge-readout screens.

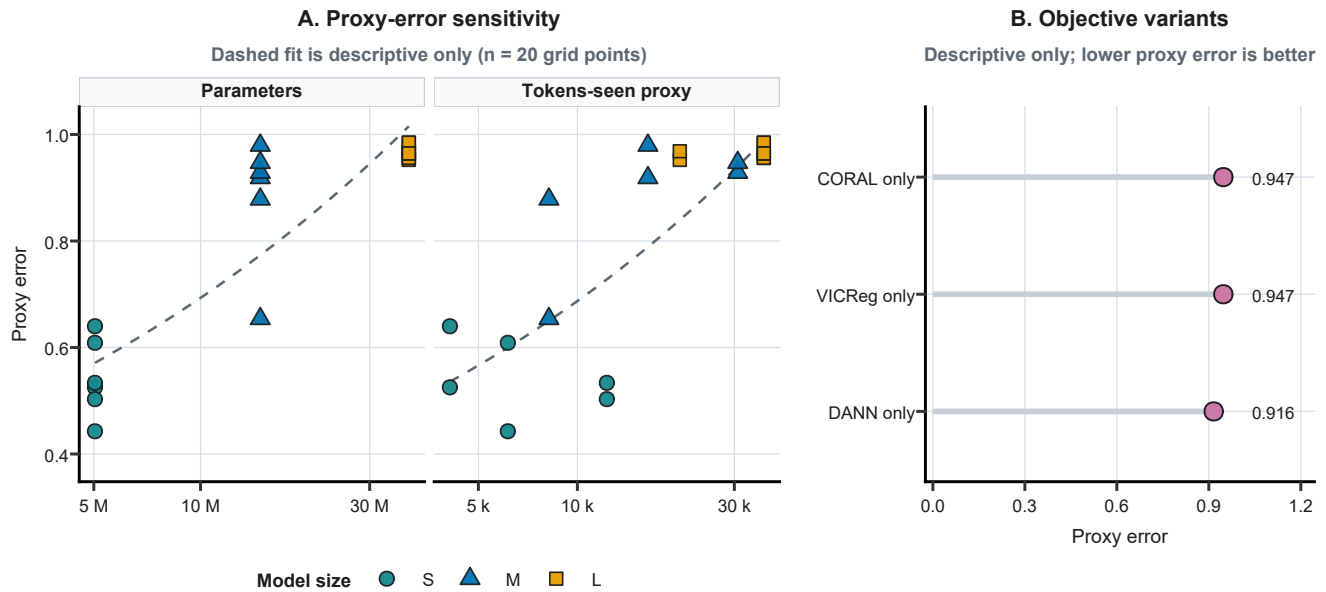


Fig. S2. Descriptive frozen-grid size-sensitivity and objective-ablation summaries. Lower frozen-grid proxy error is better. The log-log fits were computed against the tokens-seen proxy and parameter count across the small, medium, and large settings evaluated in the core grid, and the objective-ablation panel reports the available frozen-ledger component runs. These summaries are descriptive and should not be read as a universal EMG scaling law, causal objective proof, or supervised cross-dataset transfer evidence.

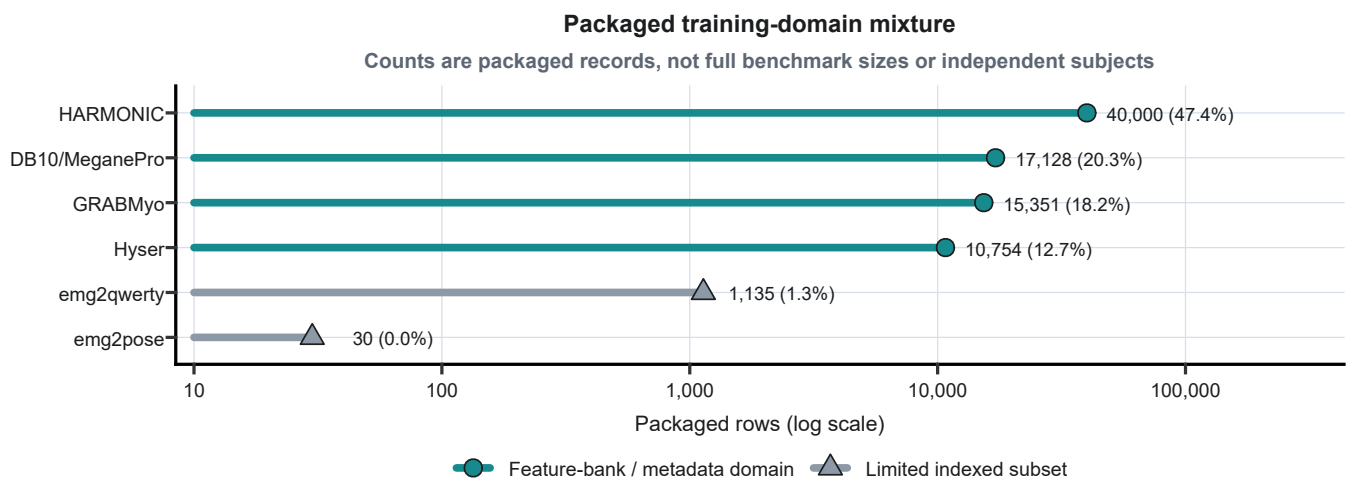


Fig. S3. Training-domain mixture used by the canonical audited package. The pretraining corpus was dominated by HARMONIC, NinaPro DB10 plus MeganePro, PhysioNet GRABMyo, and PhysioNet Hyser, with emg2qwerty and emg2pose contributing much smaller limited indexed subsets. These limited indexed subsets should not be interpreted as full emg2qwerty or emg2pose benchmark evaluation.

TABLE IX
PREPARED-PACKAGE REPRODUCIBILITY LEDGER

Packaged domain	Source role	Records used	Canonical input	Main evaluation role	Claim boundary
HARMONIC	Multimodal collaboration / stress-test data	40,000	16 channels $\times$ 64 samples	Four-domain feature-bank alignment; six-label readout and six-packaged-domain fixed-probe diagnostics	Packaged records; in the rebuilt six-packaged-domain probe, HARMONIC was reconstructed from the public text/probability stream rather than raw EMG
NinaPro DB10 + MeganePro	Prosthetic grasp-intent feature bank	17,128	16 channels $\times$ 64 samples	Four-domain feature-bank alignment; six-label readout and six-packaged-domain fixed-probe diagnostics	Combined feature-bank domain; not a standalone DB10 benchmark result
PhysioNet GRABMyo	Multi-day gesture / biometric EMG	15,351	16 channels $\times$ 64 samples	Four-domain alignment; same-dataset calibration screen; ontology gate; fixed-probe diagnostics	Transfer to Hyser blocked unless control-intent equivalence is documented
PhysioNet Hyser	HD-sEMG manipulation / neural-interface resource	10,754	16 channels $\times$ 64 samples	Four-domain alignment; same-dataset calibration screen; ontology gate; fixed-probe diagnostics	HD-sEMG topology and richer temporal structure are compressed by canonicalization
emg2qwerty	Limited indexed pretraining subset	1,135	16 channels $\times$ 64 samples	Six-label mixture and post-freeze fixed-probe diagnostics	Limited 86-file, 10-subject indexed subset in the rebuilt probe; not full emg2qwerty benchmark utilization
emg2pose	Limited indexed pretraining subset	30	16 channels $\times$ 64 samples	Six-label mixture and post-freeze fixed-probe diagnostics	Mini subset with trial/file-level splitting; too small for independent alignment or downstream claims

TABLE X
ANALYSIS PROVENANCE AND CLAIM BOUNDARY

Analysis block	Artifact status	Used for	Not used for
Canonical frozen package	Frozen before final interpretation; primary geometry, logged domain-ID, ontology, scaling, ablation, and integrity endpoints were read from saved artifacts	Primary audit case study and claim-gate ledger	Post-freeze model selection, architecture tuning, or converting failed gates into positive claims
Post-freeze domain-ID confirmation	Five independently seeded medium encoders, each evaluated with ten grouped probe seeds on all six packaged labels	Testing whether package/source identity remained recoverable from trained embeddings after the canonical readout failed near-reference calibration	Reconstructing the missing six-label adversarial head, claiming full emg2qwerty/emg2pose benchmark use, or proving physiological domain equivalence
Post-freeze downstream screens	Same-dataset session and held-out-subject ridge-readout screens with raw, pretrained, and classical signal-feature comparators	Testing whether a task-valid pretrained-feature-superiority or reduced-calibration claim passed conservative gates	Claiming online control, clinical deployment, EMGBench leaderboard performance, or universal baseline superiority
Reproducibility package	Public GitHub release, manuscript source data, source-table check-sums, and separate complete result archive	Independent tracing of figure/table values and confirmation artifacts	Redistribution of raw third-party EMG datasets or replacement of provider access terms

TABLE XI
SPLIT AND LEAKAGE CONTROLS BY ENDPOINT

Endpoint	Split unit	Leakage control	Boundary
Frozen representation package	Subject-wise validation splits where subject identifiers were available; mixed-domain self-supervised batches	Subject-overlap and forbidden-token audit tables report zero overlap or sentinel hits for the frozen subject-held-out partitions	Self-supervised pretraining is not itself a clinical test split and does not establish task-valid control transfer
Same-subject cross-session screen	Earliest session as source and later sessions as target within each subject	Calibration and evaluation were within a subject across sessions; this intentionally probes session shift, not subject independence	Not evidence of new-subject generalization
Held-out-subject few-shot screen	Held-out subject as target; source data from other subjects in the same dataset	Calibration windows and test windows for the held-out subject were trial-blocked where trial identifiers were available	Still an offline ridge-readout screen rather than online control
Fixed-embedding package/source-ID probes	Grouped logistic-regression probe seeds on frozen embeddings; subject-level where available and trial/file-level for emg2pose	Shuffled training-label controls and prediction-level confusion/support summaries were produced for all six packaged labels	Does not reconstruct the missing adversarial readout or create full emg2qwerty/emg2pose benchmark claims
Artifact-integrity audit	Saved source tables, manifests, recomputation checks, and checksum ledgers	Subject-overlap, feature-sentinel, permutation-margin, and recomputation-consistency checks are included in source data	Aggregate checks do not expose every raw trial ID because raw third-party data are not redistributed

TABLE XII
DOMAIN-IDENTIFIABILITY AUDIT LEDGER

Endpoint	Observed evidence	Missing diagnostic requirement	Audit interpretation
Logged readout, step 5	Accuracy 0.3398; entropy 1.4912; active domains: emg2qwerty, HARMONIC, GRABMyo	Held-out all-six confusion, supports, marginals, balanced accuracy, optimal relabeling, and mutual information	Aggregate mini-batch readout, not six-packaged-domain recoverability
Logged readout, step 10	Accuracy 0.1211; entropy 1.3303; active domains: emg2pose, emg2qwerty, GRABMyo	Same as above	Intermediate below-reference logged readout
Logged readout, step 15	Accuracy 0.0234; entropy 1.4749; active domains: emg2qwerty, HARMONIC, NinaPro/MeganePro	Same as above	Failed near-reference calibration check; mechanism unresolved
Post-freeze six-packaged-domain fixed probe	Five encoders $\times$ ten probe seeds recovered six packaged-domain identity with balanced accuracy 0.92262; shuffled-control balanced accuracy was 0.18829	Does not reconstruct the original adversarial readout; limited emg2pose/emg2qwerty subsets are not full benchmarks	Package/source identity remained recoverable from frozen embeddings
Remaining canonical adversarial diagnostics	No original-head six-by-six held-out confusion matrix, true supports, predicted marginals, entropy on fixed all-domain set, six-label optimal-relabel accuracy, or adversarial readout trace was exposed by the canonical artifact	Required before explaining the 0.0234 adversarial-head mechanism	The 0.0234 endpoint cannot certify invariance

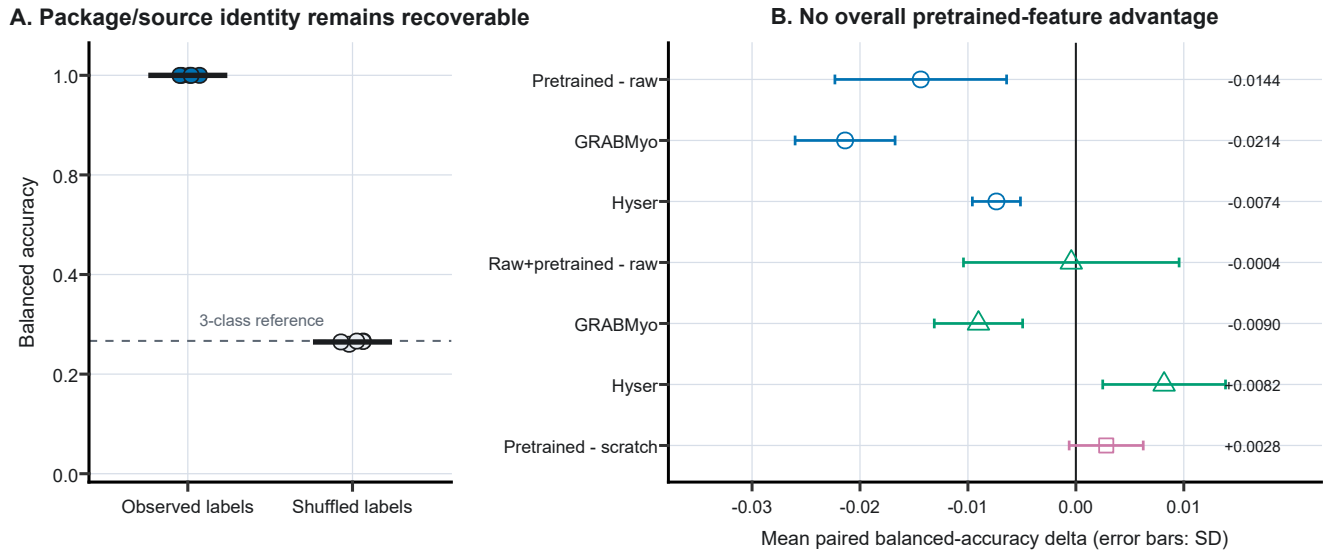


Fig. S4. Post-freeze confirmation diagnostics. (A) Independent subject-held-out package/source-ID probes across five seeded medium encoders and ten probe seeds per encoder recovered three-source identity from frozen embeddings with mean balanced accuracy 0.99989; shuffled-label controls were near the three-class reference at 0.33069. The main-text post-freeze probe table extends this recoverability diagnostic to all six packaged labels. (B) Source-plus-calibration downstream paired balanced-accuracy deltas did not establish a pretrained-feature advantage: pretrained-minus-raw was negative overall, and raw-plus-pretrained minus raw was approximately neutral overall with negative GRABMyo and positive Hyser components. Error bars show encoder-level standard deviation over the summarized cells.

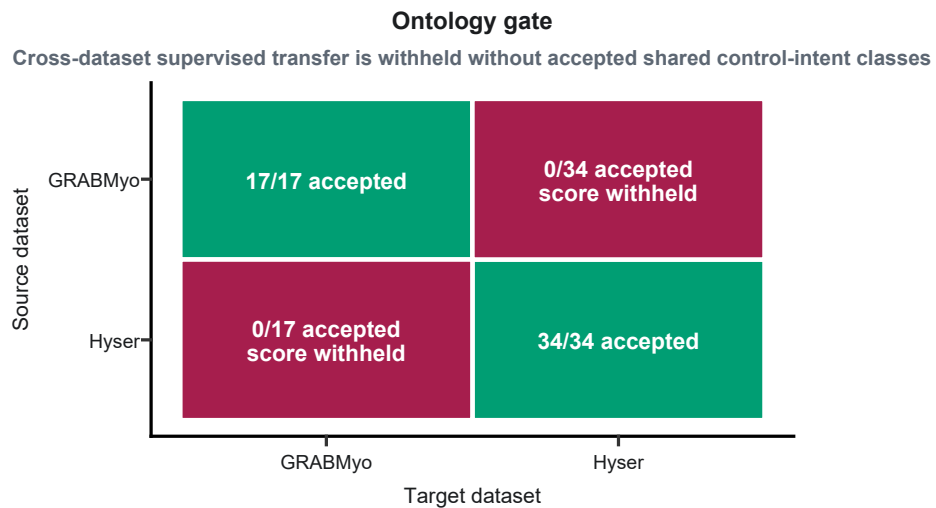


Fig. S5. Fail-closed ontology-gate summary for Hyser and GRABMyo. Because the gate found zero accepted shared control-intent classes, the package withheld a supervised transfer score rather than reporting an invalid comparison.

Check	Status	Value	Criterion
Domain-ID near chance target	FAIL	1.0000	≤ 0.05 target
Subject overlap train-test	PASS	0	count = 0
Forbidden feature tokens	PASS	0	count = 0
Permutation sanity checks	PASS	reported pass	audit logs
Recompute consistency	PASS	4.44e-16	max abs diff
Artifact completeness	PASS	reported pass	bundle complete
Artifact package hash check	PASS	reported pass	artifact archives
Off-diagonal retrieval tolerance	PASS	1.000	≥ 0.95 factor
wCKA tolerance	PASS	1.0000	≥ 0.95 factor

Fig. S6. Audit-integrity ledger for the frozen package. The near-reference target check failed, while subject-overlap, forbidden-token, permutation-margin, recomputation-consistency, off-diagonal retrieval, and wCKA tolerance checks passed. The wCKA tolerance check indicates non-catastrophic retention under the package threshold, not an improvement in wCKA.